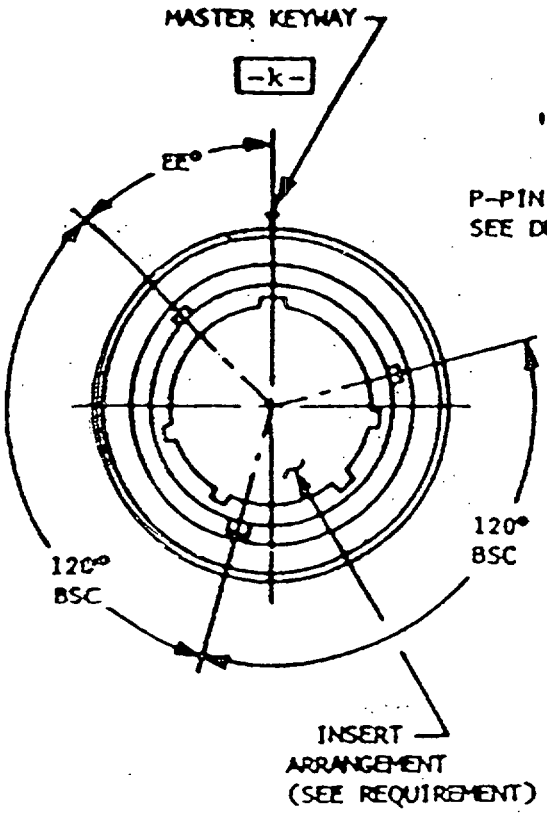
SUPERSEDING
MIL-C-0081511/52 (NAVY)
18 OCTOBER 1971

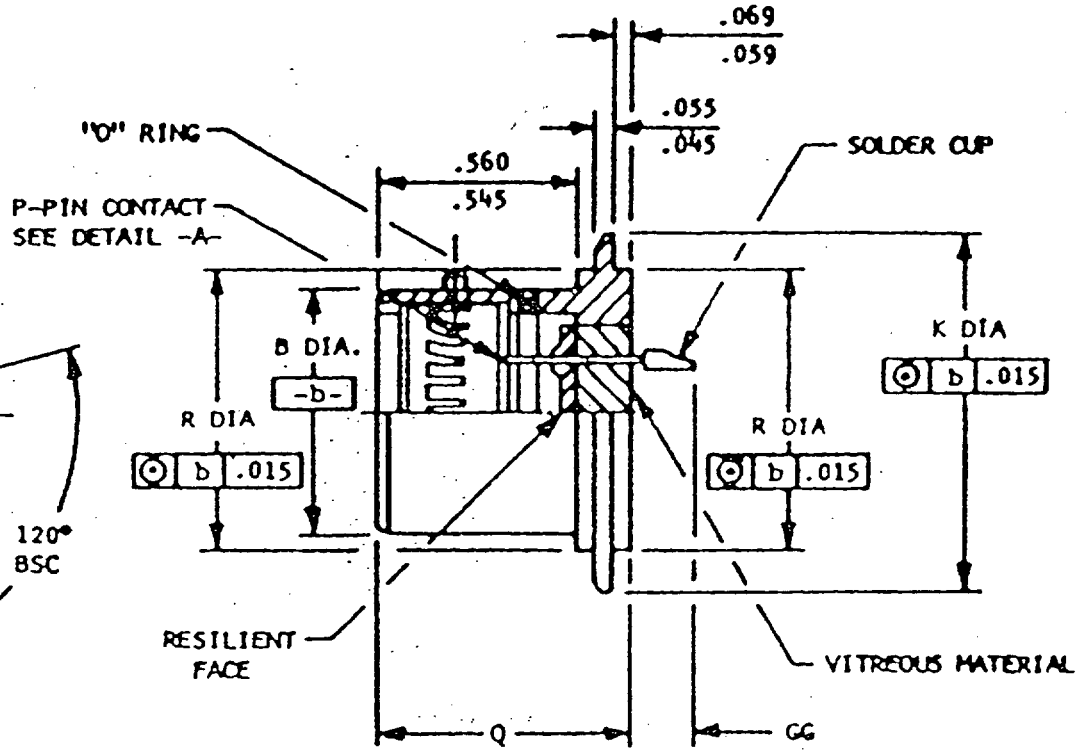
MILITARY SPECIFICATION SHEET

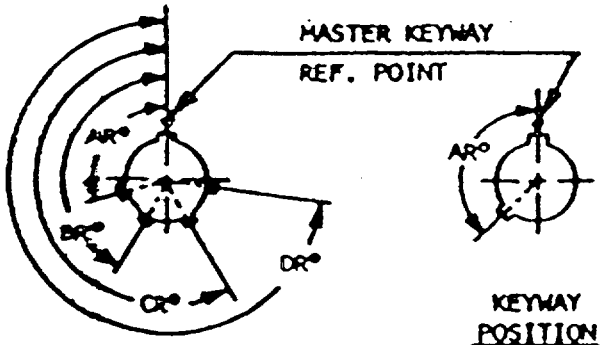
CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING: RECEPTACLE, SOLDER MOUNT, SOLDER-TYPE CONTACTS. PINS ONLY, CLASS D (SERIES 4).

THE COMPLETE REQUIREMENTS FOR PROCURING THE CONNECTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE OF SPECIFICATION MIL-C-0081511.

THIS SPECIFICATION IS MANDATORY FOR USE BY ALL ACTIVITIES OF THE DEPARTMENT OF THE NAVY.

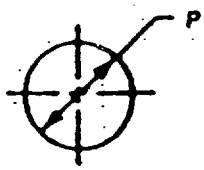


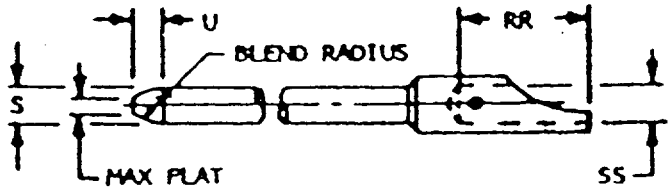




KEYWAY POSITION
NORMAL(1)POSITION
SHELL SIZE 10 AND LARGER
(SEE NOTE 5)

KEYWAY POSITION
NORMAL(1)POSITION
SHELL SIZE 8
(SEE NOTE 5)





DETAIL -A-

INCHES	MM
.015	.38
.045	1.14
.055	1.40
.059	1.50
.069	1.75
.545	13.84
.560	14.22

CONTACT SIZE	S DIA	MAX FLAT	U	RR	SS
23-22	.0275 (.70) .0260 (.66)	.010 (.25)	.022 (.56) .015 (.38)	.135 (3.45) .115 (2.92)	.038 (.97) .035 (.89)
20-20	.0410 (1.04) .0390 (.99)	.016 (.41)	.026 (.66) .018 (.46)	.135 (3.43) .115 (2.92)	.052 (1.32) .044 (1.12)
16-16	.0630 (1.60) .0610 (1.55)	.025 (.64)	.026 (.66) .018 (.46)	.185 (4.70) .165 (4.19)	.077 (1.96) .069 (1.75)
12-12	.0950 (2.41) .0930 (2.36)	.062 (1.57)	.026 (.66) .018 (.46)	.185 (4.70) .165 (4.19)	.125 (3.18) .114 (2.90)

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TABLE I

SHELL SIZE	B DIA.	F DIA.	P DIA.	Q	R DIA.	GG CONTACT SIZE 23-22	20-20	16-16	12-12
8	.553 (14.05)	.945 (24.00)	.666 (16.92)	.710 (18.03)	.653 (16.59)	.240 (6.10)	.260 (6.60)	.320 (8.13)	.420 (10.67)
	.547 (13.89)	.930 (23.62)	.656 (16.66)	.690 (17.53)	.643 (16.33)	.210 (5.33)	.230 (5.84)	.290 (7.37)	.390 (9.91)
10	.678 (17.22)	1.070 (27.18)	.791 (20.09)	.710 (18.03)	.778 (19.76)	.240 (6.10)	.260 (6.60)	.320 (8.13)	.420 (10.67)
	.672 (17.07)	1.055 (26.80)	.781 (19.84)	.690 (17.53)	.768 (19.51)	.210 (5.33)	.230 (5.84)	.290 (7.37)	.390 (9.91)
14	.928 (23.57)	1.320 (33.53)	1.041 (26.44)	.710 (18.03)	1.028 (26.11)	.240 (6.10)	.260 (6.60)	.320 (8.13)	.420 (10.67)
	.922 (23.42)	1.305 (33.15)	1.031 (26.19)	.690 (17.53)	1.018 (25.86)	.210 (5.33)	.230 (5.84)	.290 (7.37)	.390 (9.91)
16	1.054 (26.77)	1.446 (36.73)	1.167 (29.64)	.710 (18.03)	1.154 (29.31)	.240 (6.10)	.260 (6.60)	.320 (8.13)	.420 (10.67)
	1.048 (26.62)	1.431 (36.35)	1.157 (29.39)	.690 (17.53)	1.144 (29.06)	.210 (5.33)	.230 (5.84)	.290 (7.37)	.390 (9.91)
18	1.178 (29.92)	1.570 (39.88)	1.291 (32.79)	.740 (18.80)	1.278 (32.46)	.210 (5.33)	.230 (5.84)	.290 (7.37)	.390 (9.91)
	1.172 (29.77)	1.555 (39.50)	1.281 (32.54)	.720 (18.29)	1.268 (32.21)	.180 (4.57)	.200 (5.08)	.260 (6.60)	.360 (9.14)

SHELL SIZE	AR° BSC	BR° BSC	CR° BSC	DR° BSC	KEY POSITION (EE° BSC) (DEGREES) 1	2	3	4	5	6
8	128°				40	40	40	40	40	40
10	105°	140°	215°	265°	60	46	46	60	89	60
14	105°	140°	215°	265°	60	46	46	60	89	60
16	105°	140°	215°	265°	50	50	50	50	50	50
18	105°	140°	215°	265°	50	50	50	50	50	50

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH 25.4 MM.
4. DIMENSIONS ARE AFTER PLATING.
5. NORMAL MATING KEYWAY POSITION SHOWN. FOR ALTERNATE KEYWAY POSITIONS SEE MIL-C-0081511, FIGURE 4.
6. FOR SHELL SIZES 20, 22 AND 24 USE MIL-C-0081511/42 CONNECTORS.

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REQUIREMENTS:

DIMENSIONS AND CONFIGURATION: SEE FIGURE AND TABLE I.

SERIES: 4.

CLASSES: D.

SHELL SIZE: SEE TABLE II.

INSERT ARRANGEMENT: SEE SUPPLEMENT FOR MS SHEET CORRESPONDING TO DESIRED SHELL SIZE.

CONTACTS: STYLE P. SOLDER TYPE; CUP TERMINATION.

(A) CONTACT MATERIAL (SIZES 20 AND 23 ONLY): SIZE 20 AND 23 PIN CONTACTS SHALL BE MADE FROM HIGH STRENGTH NON-FERROUS ALLOY.

(A) CONTACT BEND STRENGTH: PIN CONTACTS SHALL WITHSTAND THE FORCES SPECIFIED IN TABLE III, APPLIED PERPENDICULAR TO THE CONTACT AXIS, AND SHALL NOT DISPLACE BEYOND THE LIMITS SPECIFIED.

SHELL POLARIZATION: 1 THRU 6 KEY POSITION

WEIGHT: SEE TABLE II

MARKING: PART NUMBER ON SHELL

MATING COUNTERPART: PLUG CONNECTORS SPECIFIED IN MIL-C-81511/6, /34, AND MIL-C-0081511/56.

PART NUMBER: THE PART NUMBER SHALL CONSIST OF MB1511/52 TOGETHER WITH THE APPLICABLE CODES FOR CLASS, SHELL SIZE, INSERT ARRANGEMENT, CONTACT STYLE, AND KEY POSITIONS. INFORMATION CONCERNING THE FORMULATION OF THE PART NUMBER CAN BE FOUND IN MIL-C-0081511.

TABLE II

SHELL SIZE	WEIGHT (REF) (POUNDS)
18 (CODE F)	.097
16 (CODE E)	.085
14 (CODE D)	.069
10 (CODE B)	.042
8 (CODE A)	.033

(A) TABLE III

CONTACT SIZE	MAX. DISPLACEMENT, IN.	FORCE, LBS.
23	.032	6
20	.032	10
16	.062	15
12	.062	20

CUSTODIAN:
NAVY-AS

PREPARING ACTIVITY:
NAVY-AS

REVIEW ACTIVITY:
NAVY-EC

(PROJECT NO. 5935-N929-5)

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